Laboratory of Biomedical Optics – Biophotonics

Exercise 2: Time-Resolved Photon Migration in a Semi-Infinite Medium

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1 Introduction

In Exercise 1, we modeled photon migration in an idealized infinite medium. In clinical practice (such as functional near-infrared spectroscopy (fNIRS) or diffuse optical tomography (DOT)), light is delivered and detected at the tissue surface. The tissue must therefore be modeled as a *semi-infinite medium* occupying the half-space $z \geq 0$, bounded by air ($z < 0$) across the physical interface at $z = 0$.

Refractive Index Mismatch and Internal Reflection

Photons reaching the boundary $z = 0$ attempt to escape into the air. Because tissue is optically denser than air ($n_{\text{tissue}} \approx 1.4 > n_{\text{air}} = 1.0$), light undergoes *Fresnel reflection*. Photons hitting the boundary at angles exceeding the critical angle $\theta_c = \arcsin(1/n) \approx 45.6^\circ$ undergo total internal reflection, while smaller angles are partially reflected back into the tissue.

We balance the escaping outward flux J_- with the reflected inward flux J_+ with the *Partial Current (Robin) Boundary Condition*:

$$\Phi(x, y, 0, t) - 2AD \left.\frac{\partial \Phi(x, y, z, t)}{\partial z}\right|_{z=0} = 0 \tag{1}$$

The dimensionless parameter A quantifies the index mismatch:

$$A = \frac{1 + R_{\text{eff}}}{1 - R_{\text{eff}}} \tag{2}$$

where R_{eff} is the effective diffuse reflection coefficient ($A = 1$ for index-matched media, and $A \approx 3.25$ for tissue with $n = 1.4$). In practice, $A(n_{\text{rel}})$ is evaluated via empirical polynomials (Groenhuis et al., Contini et al.).

The Extrapolated Boundary Condition (EBC)

Linear extrapolation of the fluence gradient $\frac{\partial \Phi}{\partial z}$ at the boundary into the region $z < 0$ indicates that the fluence reaches zero at a virtual plane located at distance z_e:

$$z_e = 2AD \tag{3}$$

Under the *Extrapolated Boundary Condition (EBC)*, the mixed Robin condition at the physical surface $z = 0$ is simplified to an exact Dirichlet boundary condition on the virtual plane:

$$\Phi(x, y, z = -z_e, t) = 0 \tag{4}$$

Method of Images and Analytical Solution

To solve the diffusion equation in $z \geq 0$ satisfying the boundary condition $\Phi(x, y, z = -z_e, t) = 0$:

1. **Collimated source representation**: A laser pulse perpendicularly incident at $(0, 0, 0)$ penetrates an average depth of one transport mean free path before multiple scattering fully randomizes its direction. It is modeled as an isotropic source at depth $z_s = 1/\mu_s'$:

$$\mathbf{S}_+ = \left(0, 0, \frac{1}{\mu_s'}\right) \tag{5}$$

2. **Negative image source (sink)**: To enforce $\Phi(z = -z_e, t) = 0$, an anti-symmetric negative image source is placed symmetrically opposite the extrapolated plane at depth $z_m = -z_s - 2z_e = -1/\mu_s' - 4AD$:

$$\mathbf{S}_- = \left(0, 0, -\frac{1}{\mu_s'} - 4AD\right) \tag{6}$$

3. **Superposition**: The fluence inside the medium ($z \geq 0$) is the sum of the positive and negative source Green's functions: $\Phi(\mathbf{r}, t) = \Phi_\infty(\mathbf{r}, \mathbf{S}_+, t) - \Phi_\infty(\mathbf{r}, \mathbf{S}_-, t)$, where Φ_∞ is the infinite-medium Green's function from Exercise 1.

4. **Surface flux J_z**: An optical detector placed on the surface at $z = 0$ does not measure the isotropic internal fluence rate Φ. Instead, it collects the outward-directed photon current along z (the exiting flux J_z) escaping across the boundary. By Fick's law, this outward current is proportional to the normal gradient:

$$J_z(r, t) = D \left.\frac{\partial \Phi(r, z, t)}{\partial z}\right|_{z=0} \tag{7}$$

Using the Robin boundary condition $\Phi(r, 0, t) = 2AD \left.\frac{\partial \Phi}{\partial z}\right|_{z=0}$, we directly obtain:

$$J_z(r, t) = \frac{\Phi(r, z = 0, t)}{2A} \tag{8}$$

2 Objectives

1. Implement the semi-infinite Green's function solver `TDSemiInfinite_EBC.m`.
2. Simulate the time-resolved exiting current J_z (DTOF) and study the influence of μ_a and μ_s'.
3. Vary the source-detector distance (r) and benchmark the Diffusion Approximation against Monte Carlo simulations (`Homo_MC.mat`).
4. Visualize the 3D spatial evolution of the forward-propagating photon cloud from the source into the medium.

3 Step-by-Step Instructions

Step 1: Implement the Semi-Infinite Green's Function (`TDSemiInfinite_EBC.m`)

Create the function `TDSemiInfinite_EBC.m`. Implement the Green's function following the comments in the code to correctly manage the parameters. The function should return the exiting flux J_z.
Note. Before returning J_z, impose `y(y<0) = 0` to ensure physical positivity.

Step 2: Simulation Setup and Baseline Curve

In the script `Exercise2_semiInfinite.m`, define baseline parameters and plot the first DTOF curve.

```
n = 1.4;
dt = 5;                    % time step [ps]
N = 2000;                  % number of time points
t = (0:N-1) * dt + dt/2; % time vector [ps]
```

```
mua = 0.1;                % absorption coefficient [cm^-1]
musp = 10;                % reduced scattering coefficient [cm^-1]
SourcePos = [0, 0, 0];    % source at the surface
DetPos = [2, 0, 0];       % detector at r = 2 cm
```

Step 3: Parametric Studies

Influence of Absorption (μ_a)

- Fix $r = 2\,\text{cm}$ and $\mu_s' = 10\,\text{cm}^{-1}$.
- Vary $\mu_a \in [0,\, 0.2]\,\text{cm}^{-1}$ with step $0.01\,\text{cm}^{-1}$.
- Plot the curves using `semilogy`.

How does μ_a affect the late-time tail of the curve?

Influence of Reduced Scattering (μ_s')

- Fix $r = 2\,\text{cm}$ and $\mu_a = 0.1\,\text{cm}^{-1}$.
- Vary $\mu_s' \in [5,\, 20]\,\text{cm}^{-1}$ with unit step.
- Plot the curves using `semilogy`.

How does μ_s' affect the peak arrival time ($t_{\max}$) and curve broadening? Why do the tails have identical slopes?

Source-Detector Separation (r) and Monte Carlo Comparison

- Fix $\mu_a = 0.1\,\text{cm}^{-1}$, $\mu_s' = 10\,\text{cm}^{-1}$, and vary detector distance $r \in \{1, 2, 3, 4\}\,\text{cm}$.
- Compare analytical predictions with the benchmark Monte Carlo simulations in `Homo_MC.mat`.

For which distances and time ranges does the Diffusion Approximation match Monte Carlo? Where and why does it fail? (Hint: consider the ballistic transit time $t \approx r/v$ and the assumption of isotropic radiance).

3D Spatial Fluence Evolution (Animation)

Compute the spatial distribution of fluence on a 2D slice (xz-plane at $y = 0$) as time evolves. Use the function `drawnow`.

4 Extra questions

1. In fNIRS of the brain, why does selecting a larger optode distance ($r \approx 3$–$4\,\text{cm}$) probe the cerebral cortex rather than just superficial scalp and skull?